

ADVANCED DATABASES

Databases II

Normal

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

- 1 Object-Oriented Databases (OODB)
- 2 NoSQL Databases /
- 3 Parallel Databases /
- 4 Distributed Databases /



Outline

- 1 Object-Oriented Databases (OODB)
- 2 NoSQL Databases
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What is an OODB?

- Combines object-oriented programming and database principles.
- Stores **objects** with their methods and attributes.
- Supports complex data and inheritance.

~ 2000s

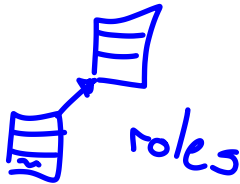
calculations
queries



What is an OODB?

- Combines **object-oriented programming** and **database principles**.
- Stores **objects** with their **methods** and **attributes**.
- Supports **complex data** and **inheritance**.

trees
graphs



OODB Features

- Encapsulation, Inheritance, Polymorphism.
- Direct object storage and identity.
- Supports OQL (Object Query Language).

```

class User {
    public grants() {
        return "Q";
    }
}

```

```

class Admin extends User {
    @Override
    public grants() {
        return "RW";
    }
}

```

serialize

```

class A {
    private b;
    public is-b() {
        return this.b
        != null;
    }
}

```

a.grants()
 a.is-b()

Demo time!



OODB Features

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Demo time!



OODB Architecture

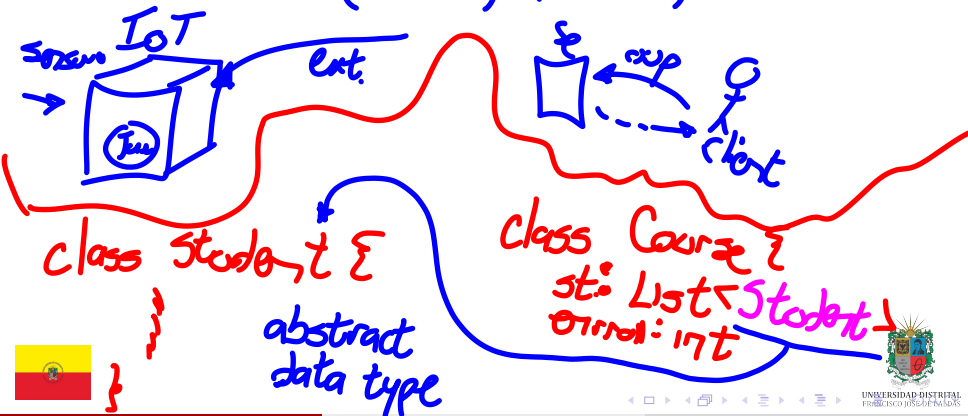
- Integrates:

JVM / MISCIL

storage

- Object-Oriented Language Runtime
- Persistence Engine
- Query Processor (OQL)

- Common deployment (embedded or client-server)



Persistence Mechanisms

- By Reachability: Root object persistence.

- By Explicit Marking: Annotate persistable objects.

- By Modification: Modified objects are persisted.

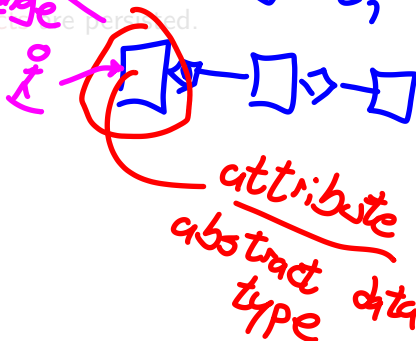
Rule of Least Knowledge

Composition,
aggregation

bad practice

A.getB() / getC();

x



Persistence Mechanisms

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- **By Modification:** Modified objects are persisted.

@save
 class A {
 b: B; ✗
 c: C; ✓
 }

class B {
 ;
 }

@save
 class C {
 ;
 }



Persistence Mechanisms

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OODB Query Example (OQL)

OQL

Student s = new Student()

```
SELECT s.name FROM Student s WHERE s.average() > 4.0
```

- Queries can follow object references.
- Better for nested, recursive, or complex types.

class Student {
 public String name;
 private double[] grades;
 public double average()
 {
 return Math.avg(grades);
 }
}

class

☐	s1
☐	s2
☐	s3
☐	s4

Demo time!



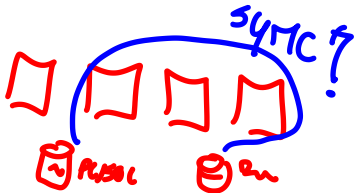
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- Better for **nested**, **recursive**, or **complex types**.

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4 Distributed Databases

No mandatory

Non-structured
Semi-structured



Types of NoSQL Databases

- **Key-Value** Redis, DynamoDB (AWS)

- Document: MongoDB, CouchDB

Python Family: Cassandra, Base

Graph: Neo4j, ArangoDB

Dictionary

• save(key, value)

• get(key)

Java

HashMap

"key1": ~

"key2": ~,

⋮

"key n": ~

S MS.

PSql → 200 MS

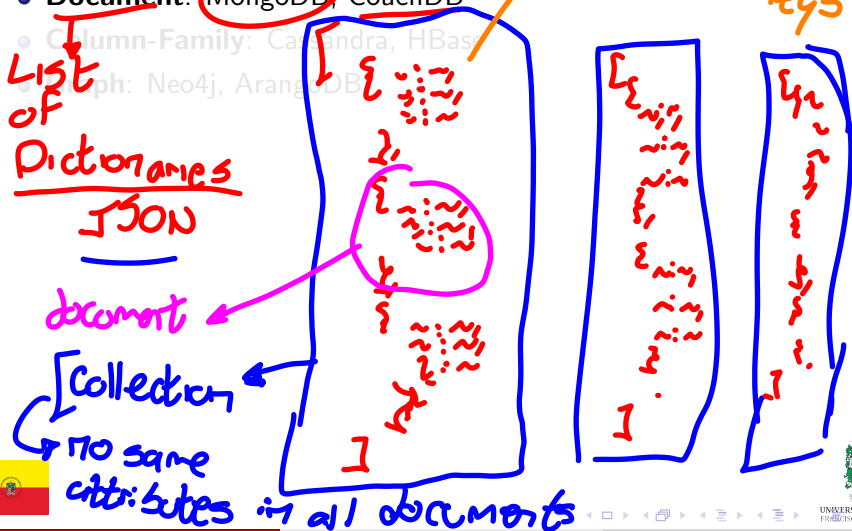


Types of NoSQL Databases

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- **Document:** MongoDB, CouchDB

↓
List
of
Dictionaries
JSON

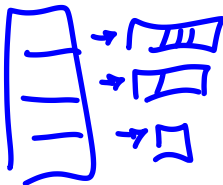
schema: common
keys



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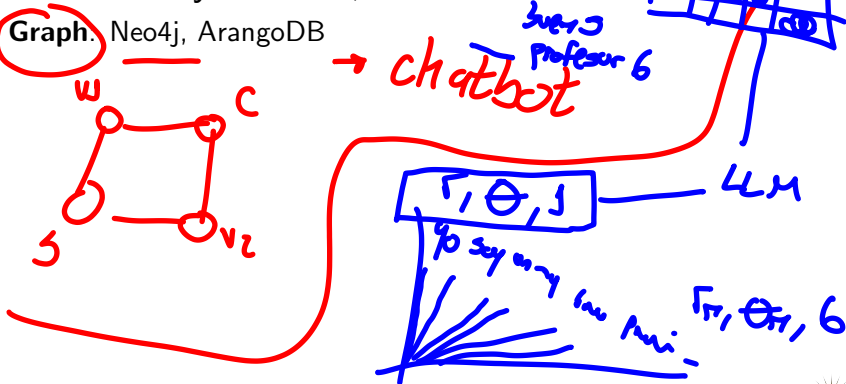
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- **Graph:** Neo4j, ArangoDB

table → column variable
↓
sparse matrix



Types of NoSQL Databases

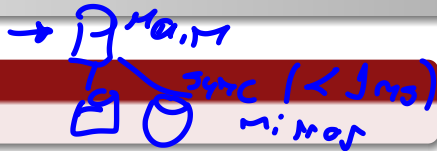
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CAP Theorem Deep Dive

Consistency (C)

Latest write is visible to all reads.



Availability (A)

System always responds.

→ no loss request

Partition Tolerance (P)

Tolerates network splits

Only two can be guaranteed at the same time.



CAP: Practical Examples

Database	CAP Preference	Comment
MongoDB	A + <u>P</u>	Eventual Consistency
Cassandra	<u>A</u> + <u>P</u>	Tunable consistency
Spanner	C + <u>P</u>	Sacrifices availability




 DataLake



 dead-times
by sync.



 clusters

 Atlas



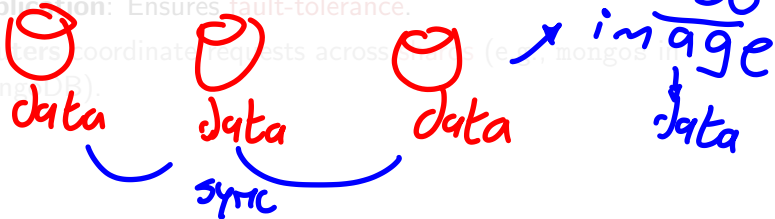
NoSQL Architecture Patterns

- **Shared-nothing architecture** enables scalability.

- **Sharding**: Distributes data across partitions.

- **Replication**: Ensures fault-tolerance.

- **Router**: Coordinates requests across shards (e.g., mongos in MongoDB).



no memory
no disk
no processing

→ Cost ↑



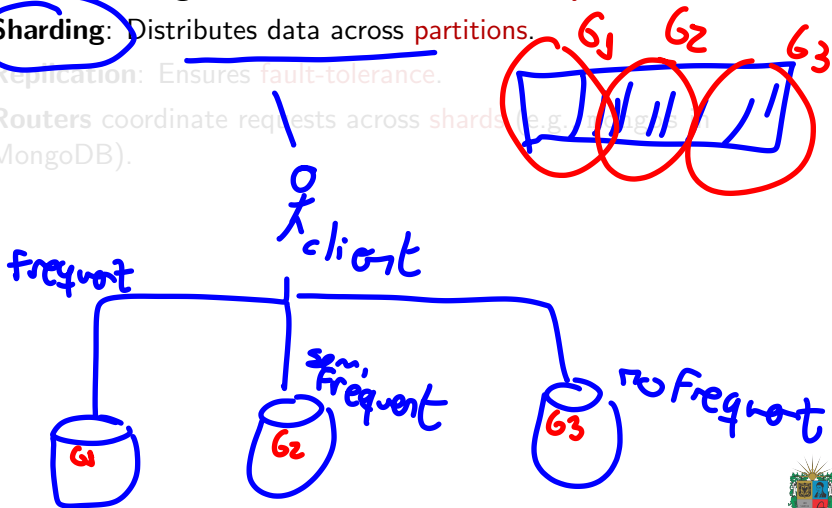
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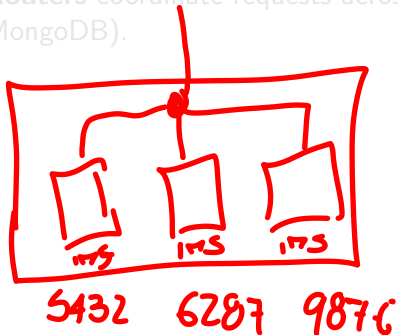
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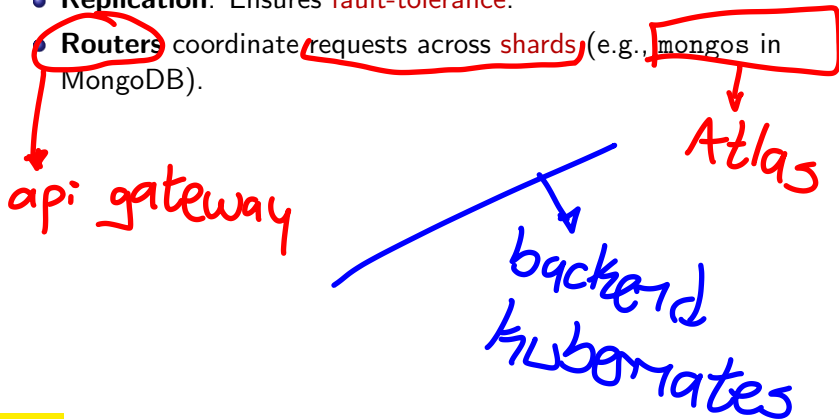
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Case Study: MongoDB vs PostgreSQL

- **MongoDB**: JSON-like schema, scalable, flexible. → *Datalake*
- **PostgreSQL**: ACID-compliant, strong relational support. → *Data warehouse*
- **Trade-offs** depend on data model complexity vs transactional needs.

Demo time!



Case Study: MongoDB vs PostgreSQL

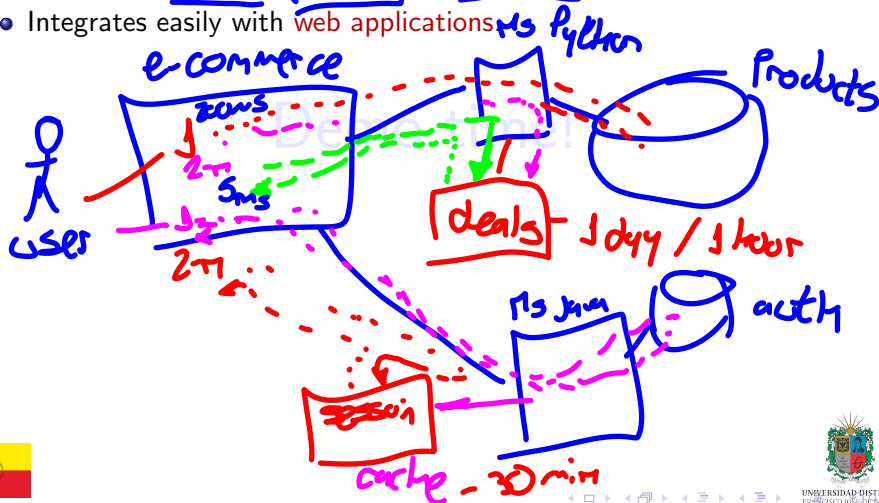
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Case Study: Redis with Web API

- In-memory key-value store.
- Excellent for caching, sessions, and real-time analytics.
- Integrates easily with web applications.



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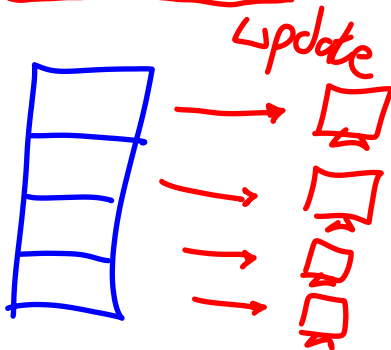
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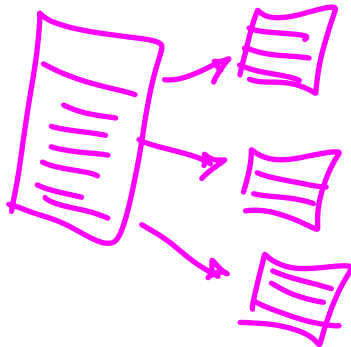
Parallel DB Concepts

- Uses multiple processors to speed up query execution.
- Exploits parallelism in data access and computation.
- Reduces query response time for big datasets.



Partitioning Strategies

- **Horizontal:** Distributes **rows**.
- Vertical: Splits columns.
- Hash Partitioning: Uniform distribution via hash.
- Range Partitioning: Based on value intervals.

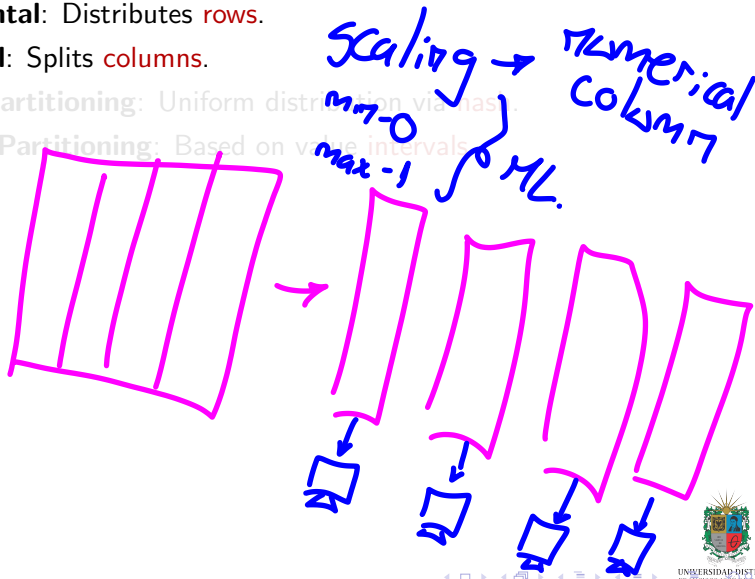


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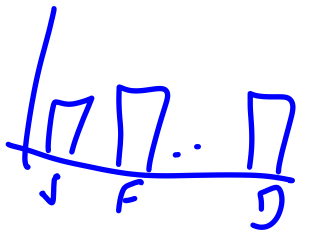
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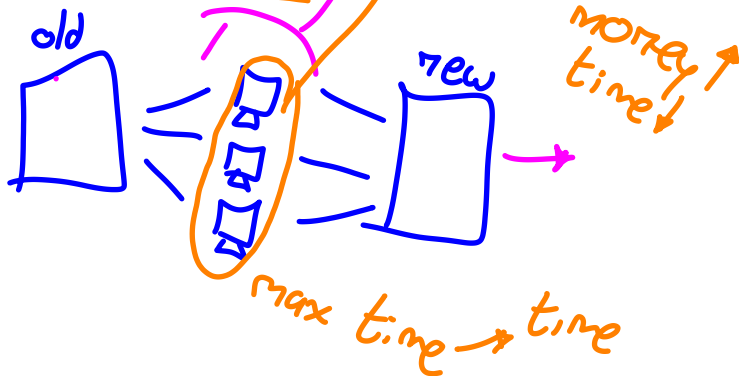
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January #
 February #
 December #



Parallel Query Cost Model

- **Startup cost (S):** Fixed overhead.
- **Communication cost (C):** Inter-node data transfer.
- **Computation cost (T):** Local processing time.
- Total: $T_{total} = S + C + T$

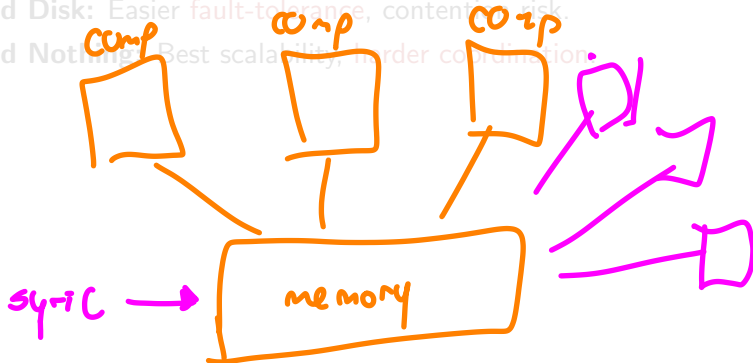


Parallel DB Architectures

- **Shared Memory:** Easy communication, poor scalability.

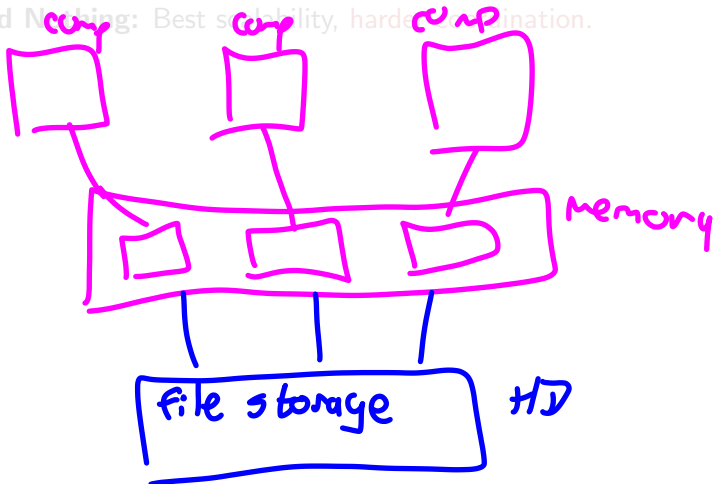
- Shared Disk: Easier fault-tolerance, contention risk.

- Shared Nothing: Best scalability, harder coordination.



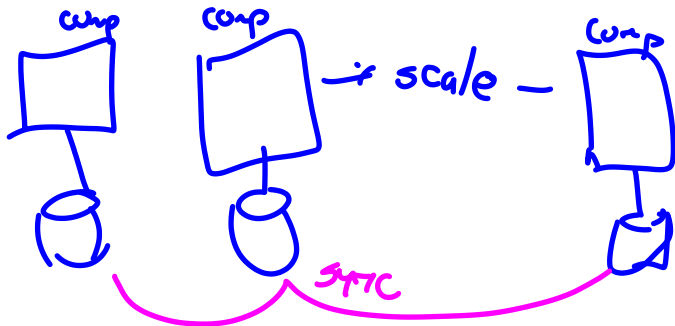
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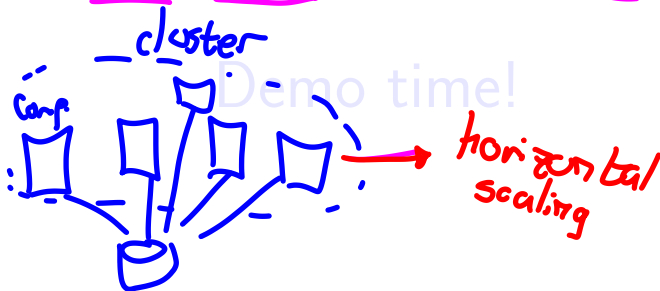
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Case Study: PostgreSQL + Citus

- **Open-source** extension for **PostgreSQL** enabling parallel, distributed queries.
- **MPP architecture** (massively parallel processing) for scaling out across nodes
- Good for **real-time analytics**, multi-tenant SaaS, and large **OLAP** workloads.
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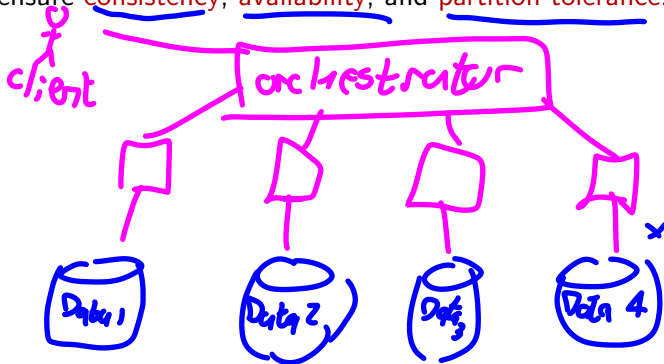
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What is a Distributed DB? /

- **Data stored** across multiple **physical nodes**.
- Appears as a **single logical database**.
- Must ensure **consistency**, **availability**, and **partition tolerance**.



Transparency Goals

- **Location** Hide physical location of data.
- **Replication**: Hide duplication.
- **Fragmentation** Hide data partitioning.

USA → data 1
EU → data 2
LATAM → data 3

NoSQL

~ ~ ~
~ ~ ~

~ ~ ~
~ ~ ~

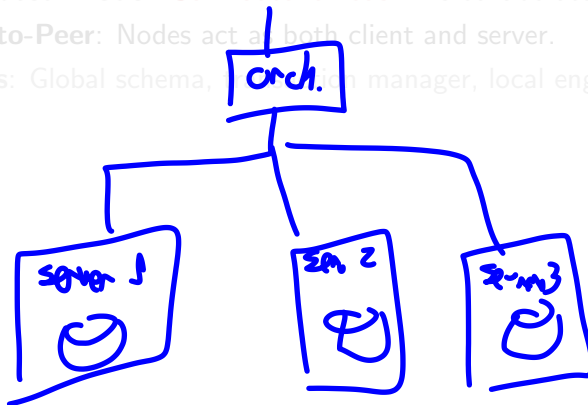
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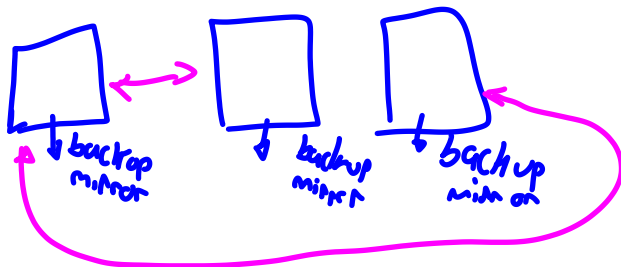
Distributed DBMS Architecture

- **Client-Server Model:** Clients query; servers store data.
- **Federated Model:** Semi-autonomous DBs collaborate.
- **Peer-to-Peer:** Nodes act as both client and server.
- **Layers:** Global schema, query manager, local engines.



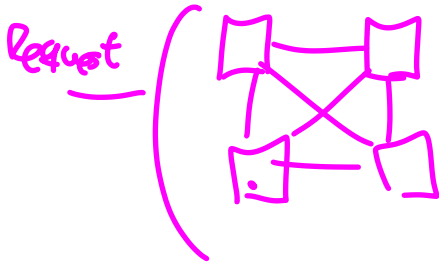
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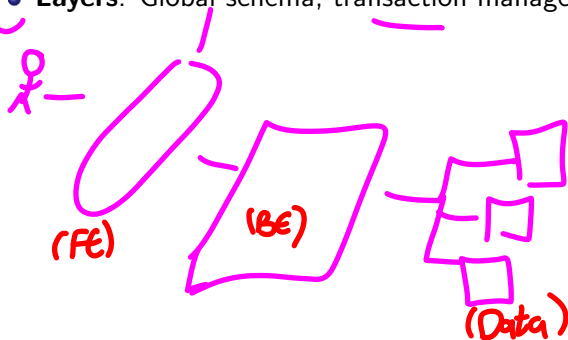
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Two-Phase Commit Protocol (2PC)

- ① **Prepare Phase:** Coordinator asks all participants to prepare.
- ② **Commit Phase:** If all vote yes, coordinator commits.

Issue: Blocking if coordinator crashes during commit.

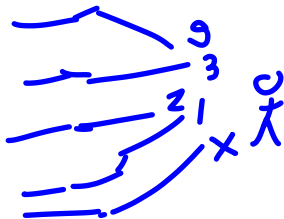
read/desired write

- Atomicity — No partial commits
- Consistent
- Events — message queues → streaming
- slow → communication/coordination



Paxos (Simplified Consensus)

- Needed for agreement in distributed systems.
- **Roles:**
 - Proposer: Suggests a value. *→ roles*
 - Acceptor: Votes.
 - Learner: Learns chosen value. *→ trust*
- Ensures consistency even if nodes fail.



Case Study: Apache Cassandra

FOSS

- Highly available and scalable NoSQL database.
- Designed for big data applications. → M-B data
- Supports multi-data center replication.
- Offers a rich query language (CQL).

→ Facebook

Demo time!



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Conclusion

internet
↓

data - str
- sen.
- int.
↑

- **Data systems** have evolved for **scalability** and **complexity**.
- Choosing the **right DB** model depends on the **workload** requirements.
- Understanding design **trade-offs** is key for architects.



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/databases-ii>

